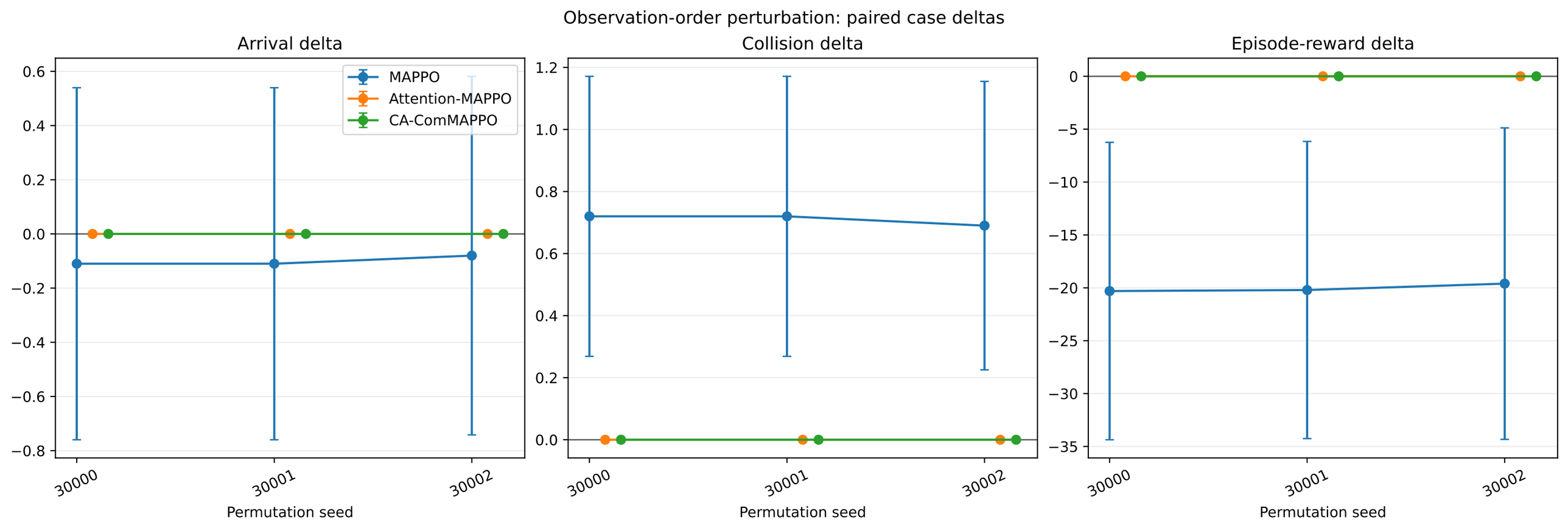
Observation-order perturbation: paired case deltas
Arrival delta
Collision delta
Episode-reward delta
MAPPO
Attention-MAPPO
CA-ComMAPPO
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
−5
−10
−15
−20
−25
−30
−35
30000
30001
30002
Permutation seed